```
import imageio.v3 as iio
import numpy as np
from scipy import ndimage
import matplotlib.pyplot as plt

# 1. BACA CITRA (imageio)
img = iio.imread("imageio:coffee.png")
if img.ndim == 3:
    img = img[..., :3].mean(axis=2).astype(np.uint8)

# 2. BUAT CITRA KONTRAS RENDAH (numpy) sebagai "citra awal"
low = (img * 0.3 + 90).astype(np.uint8)     # nilai piksel dipadatkan ke ~90-166

# 3. EKUALISASI HISTOGRAM (numpy, manual sesuai langkah di soal 2)
def equalize(image):
    hist = np.bincount(image.flatten(), minlength=256)   # langkah 1: histogram
    cdf = hist.cumsum()                                  # langkah 3: CDF
    cdf_min = cdf[cdf > 0].min()
    N = image.size
    lut = np.round((cdf - cdf_min) / (N - cdf_min) * 255)  # langkah 4: pemetaan
    lut = lut.clip(0, 255).astype(np.uint8)
    return lut[image]                                    # langkah 5: ganti piksel

eq = equalize(low)

# 4. PENAJAMAN (scipy.ndimage) - unsharp masking
blur = ndimage.gaussian_filter(eq.astype(float), sigma=2)
sharp = eq + 1.5 * (eq - blur)
sharp = np.clip(sharp, 0, 255).astype(np.uint8)

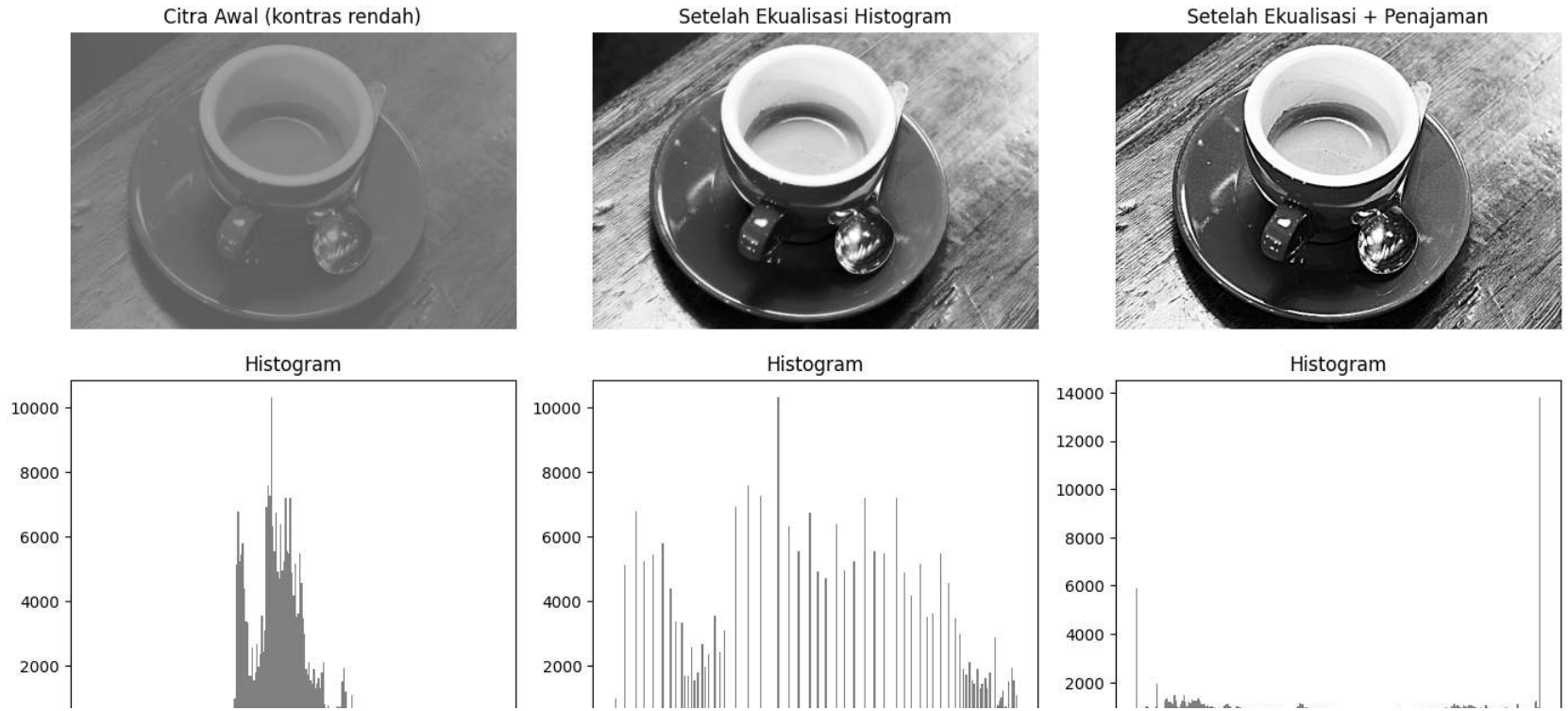
# 5. SIMPAN HASIL (imageio)
iio.imwrite("citra_awal.png", low)
iio.imwrite("citra_ekualisasi.png", eq)
iio.imwrite("citra_tajam.png", sharp)

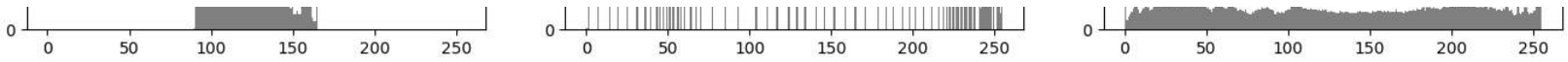
# 6. TAMPILKAN CITRA + HISTOGRAM
hasil = [low, eq, sharp]
```

```
hasil = [low, eq, sharp]
judul = ["Citra Awal (kontras rendah)", "Setelah Ekualisasi Histogram", "Setelah Ekualisasi + Penajaman"]

fig, axs = plt.subplots(2, 3, figsize=(14, 7))
for i in range(3):
    axs[0, i].imshow(hasil[i], cmap="gray", vmin=0, vmax=255)
    axs[0, i].set_title(judul[i])
    axs[0, i].axis("off")
    axs[1, i].hist(hasil[i].flatten(), bins=256, range=(0, 255), color="gray")
    axs[1, i].set_title("Histogram")
plt.tight_layout()
plt.savefig("hasil_program.png", dpi=150)
plt.show()
```

```
Imageio: 'coffee.png' was not found on your computer; downloading it now.
Try 1. Download from https://github.com/imageio/imageio-binaries/raw/master/images/coffee.png (441 kB)
Downloading: 451102/451102 bytes (100.0%)
  Done
File saved as /root/.imageio/images/coffee.png.
```




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